Baichuan Zeng (Nokk)

AI in Biomedicine | Computer Vision | Extended Reality (XR)
Born 6 February 2004 | Nationality: Hong Kong SAR

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Education

2026–Present **Elite Master's Program (M.Sc.) in AI in Biomedicine**
Technical University of Munich (TUM) | Munich, Germany

2022–2026 **B.Sc. in Computer Science**
The Chinese University of Hong Kong (CUHK) | Hong Kong SAR
Faculty of Engineering ELITE Stream student | S.H. Ho College

Research Experience

May 2026 **Final-Year Project: Prototyping a Surgical Training Platform**
CUHK | Supervised by Prof. Chi-Wing Fu
Prototyped an XR-based laparoscopic training system on Meta Quest 3; implemented tool tracking, interaction logic, UI/UX, and training scenarios for surgical skill development.

Jan 2026 **Machine Learning for TDP1 Inhibitor Prediction**
Research project
Dry-lab work: fine-tuned ChemBERTa to predict TDP1 inhibitor potency; wet-lab validation by collaborators from the CUHK Faculty of Medicine.

Mar 2025 **Participant**
Online Program in Computer Vision in Robotics
Advised by Dr. Guangming Wang (University of Cambridge); completed a review paper on AI-powered spatial intelligence in XR.

Aug 2023 **Summer Research**
Shanghai Jiao Tong University | Supervised by Prof. Zheng Zeng
Worked on the design and implementation of a morphing hybrid aerial-underwater vehicle; conducted outdoor experiments, data analysis, and paper writing.

Publications

Dec 2025 ***Fine-Tuning ChemBERTa for Predicting Inhibitory Activity Against TDP1 Using Deep Learning***
arXiv preprint | Sole author

Jun 2025 ***Cognitive UAV Tracking: Leveraging DRL and Hybrid Curriculum Learning for Target Reacquisition***
IEEE T-ASE | Second author

Apr 2025 ***Recent Advances and Future Directions in Extended Reality (XR): Exploring AI-Powered Spatial Intelligence***
arXiv preprint | Sole author

Jan 2025 ***Nezha-MB: Design and Implementation of a Morphing Hybrid Aerial-Underwater Vehicle***
IEEE International Conference on Robotics and Automation (ICRA 2025) | Third author

Honors & Awards

Jul 2025 Overseas Academic Activities Scholarship, S.H. Ho College, CUHK

Jun 2025 Summer Institute Exchange Scholarship, S.H. Ho College, CUHK

Jul 2021 Best Performance, Individual Winner, Hong Kong Mathematics Olympiad (full mark in Paper Two)

Aug 2020 Third Honor Award, Hong Kong Physics Olympiad

Technical Skills & Languages

Programming Python, C/C++, C#, Java, SQL

Tools & Platforms Unity, OpenCV, NVIDIA Isaac Sim, Gazebo, Blender

Languages English, Mandarin, Cantonese, German (A1)